National University of Computer and Emerging Sciences

Lab Manual

Data Analysis and Visualization

Lab 05

Instructor	Rida Mahmood, Asif Ahsan
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Fast School of Computing

FAST-NU, Lahore, Pakist

Objectives

★ Use OpenCV Library for Image Processing Tools

The objective of this lab is to introduce the student with OpenCV especially with respect to image processing.

References:

Lecture 8 and Lecture 9 of Theory

Some Useful Commands:

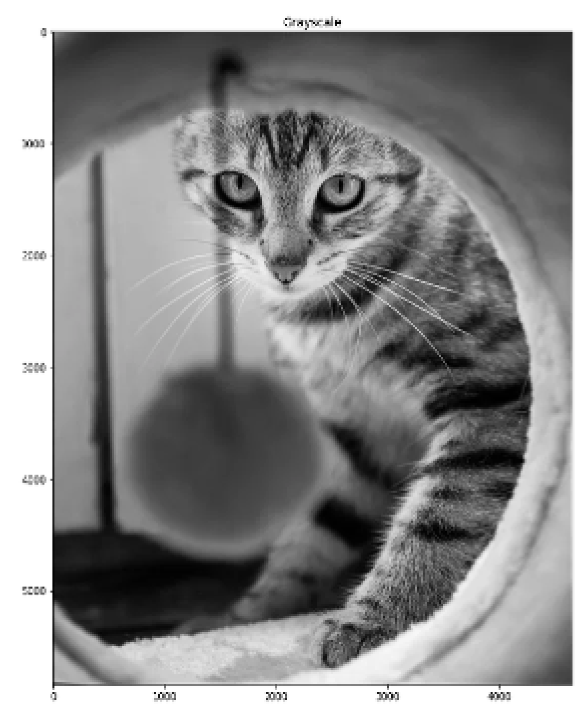
1. To create a 2D array of zeros using NumPy: my_array = numpy.zeros ((row, columns))
2. To create a 2D array of ones using NumPy: my_array = numpy.ones ((row, columns))
3. To check the size of a 2D array: size = numpy.shape(my_array)
4. Reading an image using OpenCV: my_image = cv2.imread("test_image.jpg",0) The second argument determines whether the image is read as a grayscale image or a colored image. 0 is used for reading an image as grayscale and while 1 is used for reading in color.
5. Displaying an image using OpenCV: cv2.imshow ("Title of the window", my_image). Two more commands that need to be used while displaying an image are: cv2.waitKey(x) cv2.destroyAllWindows(). The waitKey() function waits for a key being pressed for x number of milliseconds. If 0 is passed to waitKey() as an argument, it will wait indefinitely for a key press. cv2.destroyAllWindows() closes are the open image windows.
6. Writing an image to disk: cv2.imwrite("image_name.jpg", my_image)
7. Resizing an image: cv2.resize(my_image, (new_height,new_width))

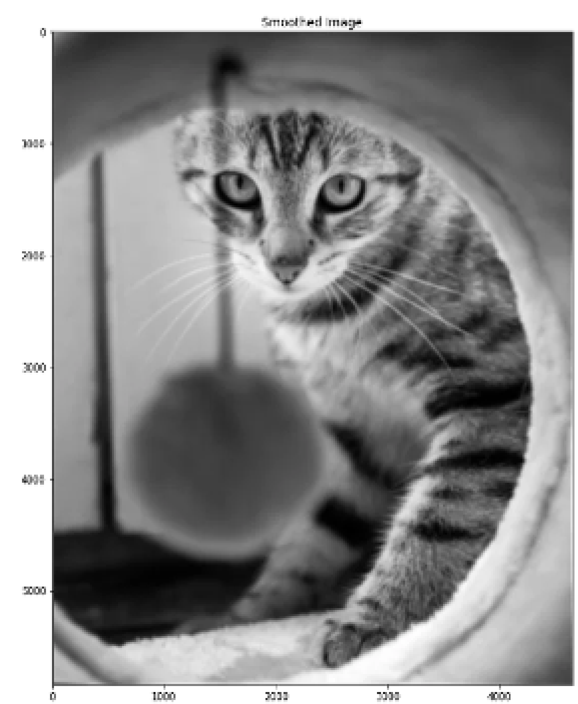
LAB TASKS

1. Design 1st order derivative and 2nd Order derivative filter and apply on input images.
2. Identify category (Gaussian or Salt and Pepper) Noise in an image and based on your assumption apply relative filter/ built-in filter on given input images.
3. Apply Logistic Regression on sample input dataset with any handcrafted features.

Image smoothing spatial filters are used for blurring and noise reduction. The output of a smoothing linear filter is simply the average of the pixels contained in the neighborhood of the filter mask. These filters are also called as averaging filters. In averaging filters, the value of every pixel in the image is replaced by the average of the gray levels in the neighborhood defined by the filter mask. This process results in an image with reduced sharp transitions in gray levels. However edges which are desirable features of an image are characterized by sharp transitions in gray levels. Hence averaging filters have the undesirable effect that blurs the edges. A major use of averaging filters is in the reduction of irrelevant detail in an image. Irrelevant details refer to the pixel regions that are small with respect to the size of the filter mask. A spatial averaging filter in which all coefficients are equal is sometimes is called a box filter.

4. Apply Noise removal filter (Smoothing/ LowPassFilter) on images and then apply logistic regression.

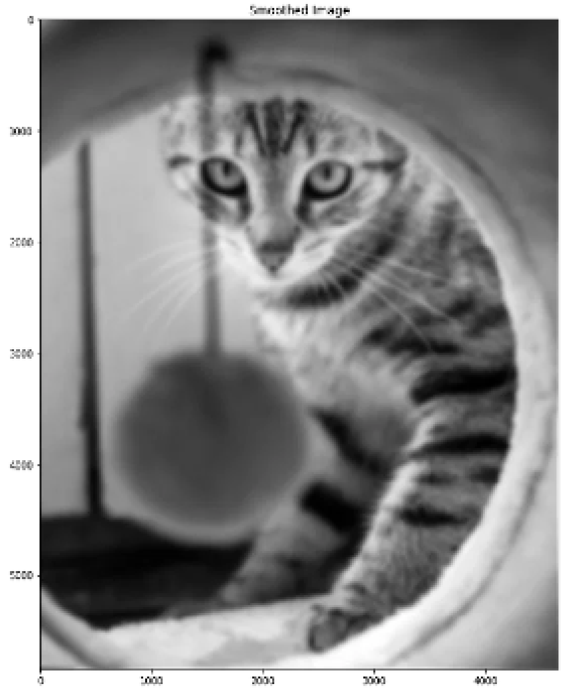


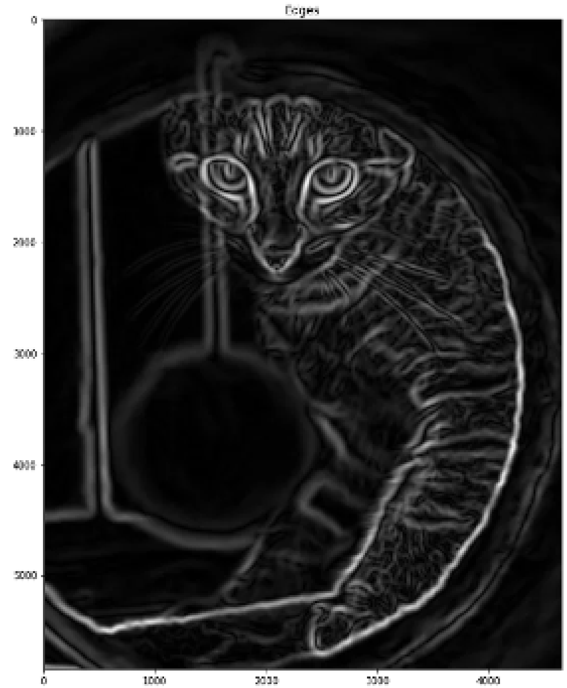


Sharpening spatial filters seek to highlight fine detail by removing blurring from images and highlighting the edges. Sharpening filters are based on spatial differentiation. Differentiation measures the rate of change of a function. It's just the difference between subsequent values and measures the rate of change of the function. The second order derivative simply takes into account the values both before and after the current value. The 2nd derivative is more useful for image enhancement than the 1st derivative due to the stronger response to fine detail and simpler implementation. The first sharpening filter we will look at is the Laplacian filter, one of the simplest sharpening filters. Applying the Laplacian to an image we get a new image that highlights edges and other discontinuities. The result of Laplacian filtering is not an enhanced image. We have to do more work in order to get our final image. We have to subtract the Laplacian result from the original image to generate our final sharpened enhanced image. In the final sharpened image edges and fine detail are much more obvious. The same process can be repeated for Sobel and Prewitt filters also.

The basic idea behind **edge detection** is to find places in an image where the intensity changes rapidly, using one of the following criteria a) Find places where the first derivative of the intensity is greater in magnitude than a specified threshold. b) Find places where the second derivative of the intensity has a zero crossing The function edge in the MATLAB provides several edge estimations based on the above criteria. Sobel edge detector computes the gradient by using the disrete differences between the rows and columns of a discrete neighborhood where the center pixel is in each row and column is weighed by 2 to provide smoothing. Prewitt edge detector finds the edges using the prewitts approximation to the first order derivatives implemented using prewitt mask. Robert edge detector finds the edges using the Robert approximation to the first order derivatives implemented using the Robert cross gradient mask. The lapalcian of Gaussian edge detector finds edges by looking for zero crossings after filtering the input image with a LoG filter. The zero Crossing detector finds the edges by looking for zero crossings after filtering the input image with a specified filter. The canny edge detector finds edges by looking for local maxima of the gradient of the input image. The gradient is calculated using the derivatives of a Gaussian filter. The method uses two thresholds to detect strong and weak edges in the output, and includes the weak edges only if they are connected to the strong edges. Therefore this method is more likely to detect true weak edges. The program uses all the above edge detectors to find the edges of a checkerboard image.

5. Apply Edge Detection (High pass) filter on images and then apply logistic regression.





6. Try a combination of above given techniques and state your shortcomings.

NOTE: You can match your results from lecture slides. Results may vary according to pictures